

CS 502P: Basic Data Science Practicum

Course Code	CS 502P
Course Name	Basic Data Science Practicum
L-T-P-C	1-0-3-3
Prerequisites	Consent of Teacher
Intended for	MSc/PhD in Chemistry, Biology, Physics, MA in HSS. Not open to B.Tech.
Distribution	Free elective
Approval	22nd Senate (Jun 2019)

Course Contents

- Introduction to Data Science; model of computation; program = data structures + algorithms
- Expressions: scalar variables, operators, precedence, data types
- Decisions: if-else; nested decisions; flow-charts
- Top-down and bottom-up program design; iterations – definite and indefinite; arrays
- Functions and modules; stats, rand, matplotlib modules; Review of statistics; Visualisation of data
- Collections: list, set, dict, Numpy array; File I/O; exceptions
- CSV file format; Pandas dataframe; Regression and interpolation; measures of goodness of fit; visual inspection – boxplots
- Data collection; cleansing; formats; units
- Case studies from experimental chemistry, biology and allied areas

Laboratory Modules

- The Python programming environment (OS and IDE); “Hello world” in Python
- Python as a calculator; input/output of numbers; type conversion
- Classification using if-else; debugging using print, breakpoints, execution of selected statements
- Trying it out using the console; flow-charts to code; initialising an array; filling an array with input numbers; printing an array
- Filling an array with random numbers; distributions; Computing statistics of an array of numbers; Plotting line, scatter, bar, histograms using matplotlib
- Use of list, set and dict; Reading data from a file into a Numpy array
- Reading .csv into a Pandas dataframe; Fitting curves to given datasets
- Gridding of datasets; conversion of file from one format to another

Text Books

1. Michael Dawson, **Python Programming for the Absolute Beginner**, 3rd Edition, Premier Press, 2003 (Chapter 1-7)
2. Jake Vanderplas, **Python Data Science Handbook**, O’ Reilly, 2016 (Chapters 1-4)

References

1. Montgomery & Runger, **Applied Statistics and Probability for Engineers**, 3rd Edition, Wiley, 2003
2. Jose Unpingco, **Python for probability, statistics and machine learning**, Springer.
3. Ben Stephenson, **The Python Workbook: A Brief Introduction with Exercises and Solutions**, Springer, 2014

The course is a subset of IC152. IC152 is intended for BTech students and assumes a strong background in mathematics and familiarity with computers. CS502P is not open to BTechs, it assumes no knowledge of computers and only 10th standard mathematics.